

Gridiron: A Market-Anchored Correlated Monte Carlo Model for Fantasy Football Projection

Design, empirical calibration, and a held-out season

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Abstract

We describe a simulation model for American football that produces joint distributions over player fantasy scoring, rather than point estimates. Three design choices distinguish it. First, scoring is sampled *per game* — one margin and one total per fixture — so that teammates and opponents share the game state that generates their statistics, and correlations arise from the structure rather than being imposed afterwards. Second, team totals are anchored to closing betting-market lines, and every candidate adjustment is tested for whether the market has already priced it; adjustments that it has are discarded as double counting. Third, all effect sizes are estimated from data rather than asserted, and effects that fail to separate from noise are reported as zero and retained in the codebase as negative results.

We estimate environmental effects on 3,341 outdoor team-games and 12,787 field-goal attempts, and matchup effects out of sample on 56,554 player-weeks. We find that wind is *not* priced into closing totals (-1.83 points per 10 mph, significant) while temperature is; that altitude has no measurable effect on field-goal accuracy once attempt distance is controlled (-0.01 yards per 1,000 ft, $se = 0.028$); and that team-versus-team history carries no information beyond the closing spread ($r = 0.026$), whereas a defence’s history against a *position* does (β between 0.16 and 0.27, all significant).

On a strictly held-out 2025 season — 272 games and 5,237 player-weeks, with all results and statistics from that season hidden from every estimator — the model matches the closing line on game outcomes (margin MAE 9.72 against the market’s 9.72) without beating it (50.7% against the spread), and is well calibrated on win probability (Brier 0.212 against a 0.249 base rate, calibration error 0.023). Player-level rank correlation with realised scoring is 0.48 overall and 0.67 for running backs. We also document two defects: projection intervals are too narrow, containing 69.3% of outcomes at a nominal 80%, and a residual under-projection of -0.82 points per player-week persists after adjusting for availability.

1 Introduction

Fantasy football scoring is a deterministic function of a stochastic process. Given the play-by-play record of a game, a player’s fantasy points follow mechanically from a scoring rule; all of the uncertainty lives upstream, in which plays happen and to whom. This suggests an obvious modelling strategy — simulate the upstream process and apply the scoring rule — and yet the projections that dominate public fantasy tools are point estimates, typically a single expected number per player per week.

A point estimate is the wrong object for almost every decision a fantasy manager actually makes. Choosing between two flex options is a question about which distribution is preferable given the rest of a lineup, not which mean is larger. Deciding whether to trade a consistent producer for a volatile one is explicitly a question about higher moments. Estimating a team’s championship odds requires the joint distribution across a roster, because a manager who stacks

a quarterback with his own receivers has a very different variance profile from one who does not, even at identical expected points.

This paper describes *Gridiron*, a system built around that observation. Its output is a set of simulated seasons and weeks — by default 20,000 of them — from which any functional of the joint distribution can be read.

Three commitments shape the design, and we state them at the outset because they are what the paper is really about.

Correlation should be structural, not bolted on. Two players in the same game are not independent draws. When a game turns into a shootout, both quarterbacks benefit; when a team falls behind, its running back loses carries and its receivers gain targets. Models that generate each player independently and then apply a correlation matrix are approximating a structure they could have simulated directly. We sample one margin and one total per fixture, allocate points to teams, allocate team production to players, and let the correlations fall out.

The market is a strong prior, and double counting is the failure mode. Closing betting lines aggregate more information than any freely available model will reconstruct. Anchoring to them is straightforward. The subtlety is what happens next: if the model then applies its own adjustment for a factor the market has *already* priced, it counts that factor twice and degrades a good prior. We therefore test every candidate adjustment against the residual from the closing line, not against raw outcomes, and apply only what survives. Section 4 shows this distinction changing a design decision.

Measured, including when the measurement is zero. Every coefficient reported here was estimated from data. Where an effect that practitioners believe in fails to separate from noise, we report zero and retain the negative result in the source, so that the absence of a knob is legible as a finding rather than an oversight. Two such results appear below: altitude on kicking, and head-to-head history.

2 Data

All football data comes from the **nflverse** project, which publishes cleaned play-by-play, weekly player statistics, team statistics, rosters, depth charts, injury reports and schedules with closing betting lines. Consensus draft rankings come from a public mirror of FantasyPros expert consensus. Venue metadata — coordinates, field elevation, roof type and surface for all 32 current stadiums plus international sites — was compiled for this work, as **nflverse** carries roof and surface per game but not location or elevation. Forecast weather is retrieved from Open-Meteo, which requires no credentials.

Estimation samples are stated with each result. The two largest are 3,341 outdoor team-games carrying both temperature and wind (1999–2025) and 12,787 field-goal attempts with kick distance (2014–2025).

3 Model

3.1 Game state

For each fixture g we sample a margin and a total,

$$M_g \sim \mathcal{N}(s_g, \sigma_M^2), \quad (1)$$

$$T_g \sim \mathcal{N}(t_g, \sigma_T^2), \quad (2)$$

where s_g and t_g are the closing spread and total, and the dispersions $\sigma_M = 12.73$ and $\sigma_T = 13.19$ points are the residual standard deviations of realised outcomes around closing lines, estimated on the same historical sample. Team points follow by construction,

$$P_g^{\text{home}} = \frac{1}{2}(T_g + M_g), \quad P_g^{\text{away}} = \frac{1}{2}(T_g - M_g), \quad (3)$$

clipped to $[0, 80]$.

Sampling at the level of the fixture rather than the team is the single most consequential structural choice in the model. Drawing two independent team scores permits both sides of a game to be simulated as winning, which is not a state of the world; more importantly, it destroys precisely the correlation that makes lineup construction interesting.

3.2 Usage allocation

Team production is allocated to players by Dirichlet draws over usage shares,

$$(\pi_1, \dots, \pi_K) \sim \text{Dir}(\alpha\bar{\pi}_1, \dots, \alpha\bar{\pi}_K), \quad (4)$$

where $\bar{\pi}_k$ are projected mean shares and the concentration α is fitted to observed week-to-week share variance. Separate allocations govern targets, carries and quarterback dropbacks.

Mean shares are estimated by empirical-Bayes shrinkage of each player’s history toward a positional and depth-chart prior, with explicit prior sample sizes so that a player with three games of history is pulled hard toward the prior and one with three seasons is not. Rates such as catch rate, yards per carry and interception rate receive the same treatment.

3.3 Season-level latent factors

Independent weekly sampling produces a leaderboard that is far too flat: it generates the average season repeatedly rather than the spread of seasons that actually occurs. We therefore draw, once per simulated season per player, latent multipliers on role and efficiency, together with a team-strength surprise ($\sigma \approx 2.6$ points per game) and a pass/rush touchdown split. These persist across all weeks of that simulated season.

A subtlety arises here that is easy to get wrong. A log-normal multiplier on Dirichlet concentration parameters inflates expected shares by Jensen’s inequality, drifting the entire board upward by roughly 10%. We normalise the shock by its share-weighted team mean so that expected shares are preserved.

3.4 Availability

Absences are persistent, not independent across weeks. We model availability as a two-state Markov chain per player, with transition probabilities fitted from injury-report history, so that a simulated injury produces a run of missed games rather than isolated zeros.

3.5 Market blending

Bottom-up projections are blended with consensus expert rankings by isotonic regression of model points-per-game on market rank, fitted *within each position*. Fitting across positions is incorrect: an overall draft board encodes positional scarcity, so a quarterback taken 107th is not the 107th-best scorer, and only the within-position ordering carries points information.

Table 1: Environmental effects, within-team fixed effects, 3,341 outdoor team-games. “vs. line” is the residual of realised total from the closing total — the quantity that determines whether an adjustment is additive to a market-anchored model or duplicative of it.

	per +10 mph wind	per -10°F	
Rushing share	+0.0167	+0.0071	both significant
Completion rate	−0.0152	−0.0066	both significant
Yards per attempt	−0.324	−0.082	both significant
Passing yards	−16.30	−3.57	both significant
Rushing yards	+1.71	+1.84	wind n.s.
Total vs. line	−1.83	−0.07	wind significant; cold n.s.

4 Environmental effects

4.1 Identification

Marginal comparisons of outcomes across weather conditions confound weather with the teams that play in it: cold-weather franchises differ systematically from dome franchises. We therefore estimate within-team fixed effects, demeaning both outcomes and conditions by team-season, so that the identifying variation is the same team playing in better or worse conditions than it usually does.

The second and more important choice is the dependent variable. To decide whether an adjustment should be applied *on top of* a market-anchored total, the relevant quantity is not the outcome but the residual from the closing line. An effect visible in raw scoring but absent from the residual is one the market has already priced.

4.2 Results

Table 1 reports both. Per 10 mph of wind, passing yards fall by 16.3, completion rate by 1.5 percentage points and yards per attempt by 0.32; rushing share rises by 0.017. Critically, the residual from the closing total falls by 1.83 points and is significant: *the market does not fully price wind*. Per 10°F of cold, the compositional effects are present and significant but the residual is -0.07 points and is not.

The model therefore applies the wind effect to the game total and the temperature effect to composition only. Applying temperature to the total as well would count it twice, since the closing line already reflects it.

4.3 Kicking, and a negative result on altitude

Raw field-goal percentage appears to fall with wind and then recover above 15 mph, which is an artefact of attempt selection: coaches stop attempting long kicks in difficult conditions, so the surviving attempts are easier. We therefore fit make probability conditional on distance,

$$\Pr(\text{made}) = \sigma(\beta_0 + \beta_d d + \beta_w w + \beta_t \tau + \beta_e e), \quad (5)$$

on 12,787 attempts. Results appear in Table 2.

Wind costs 1.63 yards of effective range per 10 mph. Altitude does not separate from zero. Denver’s attempts average 39.2 yards against 38.7 elsewhere, and conditional accuracy is flat in elevation. The underlying physics is not in doubt — air density at 5,280 ft is roughly 82% of sea level — but the effect on professional kickers is below the noise floor of this sample. The model applies no altitude adjustment, and the estimate is retained in the source so that its absence reads as a finding.

Table 2: Field-goal accuracy conditional on distance, 12,787 attempts (8,372 outdoor with weather). Coefficients are on the logit scale; the final column converts to equivalent yards of kick distance.

	coefficient	std. error	equivalent yards
Distance (per yard)	−0.1116	0.0038	—
Wind (per 10 mph)	−0.1817	0.0626	−1.63
Temperature (per −10°F)	−0.0186	0.0188	n.s.
Elevation (per 1,000 ft)	−0.0013	0.0282	−0.01, n.s.

Table 3: Out-of-sample predictive weight of a defence’s prior relative fantasy points allowed, by position. The fitted β is the shrinkage applied in production.

Position	n	β	std. error	r
QB	3,989	0.209	0.040	0.082
RB	3,998	0.266	0.039	0.108
WR	3,998	0.176	0.045	0.062
TE	3,979	0.163	0.041	0.063

5 Matchup effects

Two quantities are commonly called “matchup history” and they behave differently.

Team-versus-team history predicts nothing. On 5,798 games with at least two prior meetings, regressing the residual from the closing spread on the mean margin of the last six meetings gives $\beta = 0.043$ (se = 0.022), $r = 0.026$, $R^2 = 0.0007$; using only the most recent meeting gives $r = 0.003$. Neither is significant. Whatever it means for one team to “own” another, it does not survive a point spread. The system computes and displays this history, and weights it at exactly zero.

Defence-versus-position history does predict. Evaluated strictly out of sample — prior weeks only, predicting the following week — a defence’s relative fantasy points allowed to a position carries the coefficients in Table 3, all significant.

These coefficients are not chosen; they *are* the correct shrinkage. A defence that has allowed 67% more than average to tight ends should be expected to allow about 11% more next week, not 67%: regression to the mean is most of the story, and β measures how much of it. The adjustment is applied weekly only. Over a full season a team faces seventeen defences and the effect averages out, so incorporating it into a season-long draft ranking would add noise in the costume of information.

6 Held-out evaluation

6.1 Protocol

We hold out the 2025 season entirely. A cutoff mechanism caps every historical load at 2024, including explicit multi-season requests, so that usage profiles, efficiency priors, team-strength estimates and all fitted constants are computed without any 2025 observation. Schedules for 2025 remain visible — fixtures and closing lines are pre-game information and are the anchor the model is built on — but their result columns are redacted at load time, so no estimator can observe an outcome.

We regard this redaction as the essential detail. An earlier version of the protocol capped only the default “load all history” path, while `team_context` requested seasons explicitly and thereby saw the entire holdout season. That version reported a margin MAE of 6.97 against a market benchmark of 12.03 and an implied 84% against the spread — results that are impossible, and whose implausibility is what exposed the leak. We report the episode because a leaked backtest is worse than none, and because its signature was not a warning but an implausibly good number.

Weekly matchup factors are permitted to use prior weeks within the holdout season, as they would be in live use.

6.2 Determinism as a correctness check

Establishing the holdout also surfaced a class of defect we had not been looking for. Running the backtest twice on identical inputs produced different metrics — margin MAE moving in the third decimal, and the against-the-spread rate moving by six percentage points. Since the simulation is explicitly seeded, this should have been impossible, and chasing it found three genuine bugs upstream of the simulator.

First, the market board assigned integer ranks after sorting on average draft position alone. Players tied on ADP therefore took ranks in whatever order the sort happened to emit. Because the market blend is a global isotonic fit over those ranks, a permutation of three tied players moved projected points for most of the board.

Second, kickers were modelled by a dedicated routine *and* carried in the general roster pool, and the two were concatenated. Twenty-nine players appeared on the board twice, where they double-counted in every team-level aggregation.

Third, the routine that reduces each team to one kicker did so with an order-dependent deduplication. Beyond flipping between runs, it selected the wrong player: Cincinnati resolved to a kicker with no recent attempts over the incumbent starter. Selection now ranks by attempts in the previous two seasons with an identifier tiebreak.

None of these would have been caught by the test suite, which asserts properties of a single run. All three were caught by the weaker requirement that two identical runs agree. We note it because determinism is usually filed under engineering hygiene, and here it functioned as a correctness test that the statistical checks did not replicate. After the fixes the backtest reproduces bit-for-bit, and all figures below are from a reproducible run.

6.3 Game-level results

Table 4 reports performance over 272 regular-season games.

The model reproduces the closing line almost exactly on both margin and total, and does not demonstrably beat it. The 55.9% against the spread is the one figure here that might look like an edge, and it is not one: over 272 games the standard error on a coin flip is 3.0 percentage points, so this is 1.94 standard errors from chance — the kind of result that arises about one season in twenty by luck alone. A single season cannot separate a 56% bettor from a 50% one, and we do not claim to be the former. This is what a model should look like when it takes the closing line as its prior and adjusts only for effects it can demonstrate are unpriced. Win probabilities are well calibrated, with a calibration error of 0.023 and close agreement between predicted and realised frequencies in every bin below 0.8; the top bin ($n = 23$) realises 0.957 against a predicted 0.853, which at that sample size is two games.

We emphasise this null result. The system is not a betting edge on game outcomes, and reporting it as one would be false. Its value lies downstream, in allocating a well-calibrated game distribution to individual players.

Table 4: Held-out 2025 season, game level, 272 games at 6,000 simulations each. Market benchmarks use the closing line.

	Model	Closing line
Margin MAE	9.71	9.72
Margin RMSE	12.27	—
Total MAE	10.39	10.39
Total RMSE	13.18	—
Win accuracy	0.654	—
Brier score	0.212	base rate 0.249
Log loss	0.608	—
Calibration error	0.023	—
Beat the closing line	0.559	$z = 1.94$, n.s.

Table 5: Held-out 2025 season, player level. “[played]” divides the projection by simulated availability, which is the like-for-like comparison against outcomes that are observed only when the player appeared. The final column is the share of outcomes inside the nominal 80% interval.

	n	MAE	MAE played	bias	bias played	Spearman ρ	in 80% band
All	5,237	4.97	4.96	−1.65	−0.82	0.482	69.3%
QB	564	7.12	6.90	−3.46	−1.83	0.445	60.6%
RB	1,317	4.63	4.55	−2.88	−2.11	0.666	75.3%
WR	2,047	4.39	4.47	−1.27	−0.33	0.638	83.4%
TE	929	4.22	4.07	−3.54	−3.19	0.541	62.0%

6.4 Player-level results

Table 5 reports 5,237 player-weeks. Because a projection averages over simulations in which a player was injured or benched and scored zero, while an actual is observed only when he played, we report error both unconditionally and divided by the simulated active share.

Rank correlation with realised scoring is 0.67 for running backs and 0.64 for receivers, the two positions where usage is most persistent, and weakest for quarterbacks, where a small population and high touchdown variance dominate.

7 Limitations

Two defects are visible in Table 5 and we state them plainly.

Intervals are too narrow. The nominal 80% interval contains 69.3% of outcomes, and only 60.6% for quarterbacks and 62.0% for tight ends. The model understates tail risk, which means published boom and bust rates are conservative and the distribution is sharper than the world. For a system whose entire premise is that the distribution matters more than the mean, this is the most consequential weakness, and closing it should take priority over any improvement to central tendency.

Residual under-projection. After adjusting for availability, projections remain low by 0.82 points per player-week overall and by 3.19 for tight ends. Part of this is selection: the evaluation sample contains players who accumulated statistics, and preseason projections spread opportunity

across a depth chart while realised opportunity concentrates. Part is likely genuine miscalibration of touchdown rates. We have not separated the two.

Further limitations: the model has no representation of benching, which is why replacement-level quarterbacks project too favourably; weather beyond temperature, wind and forecast precipitation is ignored, and precipitation effects are a stated prior rather than a fitted coefficient because `nflverse` does not carry precipitation; kicker and defence models are deliberately simple; and the evaluation covers a single season, so all differences reported here are subject to one-season sampling variation.

8 Conclusion

The contribution of this work is less a modelling technique than a discipline. Simulating games rather than players yields correlations for free. Anchoring to market lines yields a strong prior for free. What is *not* free, and what we argue deserves more attention than it usually receives, is the question of which adjustments the prior has already absorbed. Wind survives that test; temperature does not. Positional matchup history survives it; team-versus-team history does not. Altitude, despite sound physics and firm folk belief, does not survive at any sample size available to us.

On a genuinely held-out season the resulting system matches the closing line without beating it, produces well-calibrated win probabilities, and ranks player scoring with a correlation of 0.48 overall and 0.67 for running backs. It also produces intervals that are too narrow and projections that are slightly too low. We report the negative results and the defects alongside the positive ones, on the view that a projection system whose failures are documented is more useful than one whose successes are.

Reproducibility

The system, the fitting scripts and the backtest harness are available in the project repository. The holdout is reproduced with `python scripts/backtest.py --season 2025 --sims 6000`, which sets the training cutoff, rebuilds every estimator from data through 2024, predicts all 272 games and writes the metrics reported here. All figures in Tables 1–5 are produced by that pipeline.